

Comparative Evaluation of Classical Geometric Methods for Steppable-Surface Segmentation in Simulated Staircase Point Clouds

Riyadh Lakadimu

Master Program in Electrical Engineering
Politeknik Elektronika Negeri Surabaya (PENS)
Surabaya, Indonesia
riyadhakadimu724@gmail.com

Abstract—Reliable identification of steppable surfaces is a prerequisite for stair-climbing and stair-following robots. This paper presents a controlled, fully labeled benchmark for steppable-surface segmentation built from a parametric mathematical model of a four-step staircase point cloud with LiDAR-style noise and outliers, and a systematic comparison of four classical geometric segmentation methods: orientation-gated RANSAC plane fitting, PCA-based normal analysis, orientation-aware DBSCAN clustering with slope analysis, and height-histogram analysis. On a cloud of 18,333 points, the DBSCAN-with-slope method attained the highest tread-class scores ($F1 = 0.943$, $IoU = 0.892$), followed closely by PCA normal analysis ($F1 = 0.927$). Plain RANSAC was the least precise because horizontal plane hypotheses absorb riser points at tread heights. A noise-sensitivity study shows the tread-class $F1$ degrading from 0.956 to 0.543 as the height noise σ_z grows from 0.01 m to 0.07 m. We discuss the safety implications of these error modes for robotic stair traversal and argue for ensemble fusion with safety margins. The benchmark, code, and equations are released to support reproducible comparison.

Index Terms—point cloud, segmentation, RANSAC, PCA, DBSCAN, staircase, mobile robot, traversability, LiDAR

I. INTRODUCTION

Robots that climb or follow staircases, along with assistive devices for the visually impaired, must reason about three-dimensional structure to act safely. Legged and wheeled platforms increasingly depend on egocentric depth sensing to negotiate stairs and other discontinuous terrain [1]–[3]. Shared-autonomy systems likewise demand reliable geometric cues before a foot or wheel is committed to a surface [4]. Across all of these settings the central perceptual question stays the same: which parts of an observed point cloud form a *tread*, the steppable horizontal surface on which weight can be safely placed, and which form a near-vertical *riser* or are simply spurious edge and outlier returns.

The problem is safety-critical. Labeling a tread as non-steppable only sacrifices an option, but labeling a riser or edge as steppable invites a foothold onto a vertical face, which produces slips, stumbles, or falls. The same hazard motivates stair perception for the visually impaired, where undetected steps are a recognised fall hazard behind dedicated stair-detection systems [5]. Reliable tread detection therefore

needs high recall and, just as importantly, high precision on the steppable class.

Recent work attacks stair and terrain perception with deep neural networks that learn point-wise semantics directly from raw clouds [5], [6]. These methods are powerful but data hungry. They require large volumes of densely labeled three-dimensional scans, which are costly to acquire for the specific geometry of stairs, and they offer limited interpretability. The tread sub-problem also lacks controlled, reproducible benchmarks with exact ground-truth labels, so reported results are hard to compare across studies. Classical geometric methods are fast, transparent, and label-free by contrast. Yet despite their maturity in plane segmentation and traversability estimation [7], [8], they remain under-compared on the staircase setting under matched conditions.

We address this gap with a controlled, fully reproducible study of classical geometric segmentation for steppable-surface detection on staircases. Our contributions are as follows:

- A parametric, fully labeled staircase point-cloud generator that emits exact tread, riser, and other labels while modeling LiDAR range jitter and outliers, enabling reproducible evaluation.
- A unified three-class problem formulation together with the implementation of four classical methods under a common orientation-aware pipeline: RANSAC plane fitting, PCA normal analysis, DBSCAN with slope confirmation, and a height-histogram level detector.
- A quantitative comparison of all four methods on the steppable (tread) class, accompanied by a controlled noise-sensitivity study that characterizes degradation as height noise grows.
- A safety-oriented discussion of the results for robotic stair traversal, relating precision and intersection-over-union to the risk of unsafe footholds.

The remainder of this paper is organized as follows: Section II reviews related work, Section III presents the proposed data generator and the four segmentation methods, Section IV reports the comparative evaluation and noise study, and Sec-

tion V concludes.

II. RELATED WORK

Detecting steppable surfaces in staircase point clouds touches several threads of point-cloud processing. We organise the literature thematically and weigh each line of work against the requirements of our problem: fast, interpretable, and label-free segmentation of treads and risers.

A. Geometric plane segmentation

The dominant paradigm for extracting planar structure from unorganised clouds is Random Sample Consensus (RANSAC). Li *et al.* [9] improve standard RANSAC for indoor and urban plane segmentation by guiding sampling with Normal Distribution Transformation cells, which mitigates the spurious planes that plain RANSAC fits across disjoint surfaces. The same plane-fitting machinery appears in RGB-D object recognition [10] and inside real-time planar LiDAR SLAM [11], where planes act as map primitives. Such methods are efficient and interpretable. The difficulty is that they treat planes as generic primitives with no orientation prior. On a staircase the globally dominant plane is the oblique ramp spanning all steps, so unconstrained fitting absorbs riser points into tread planes. This observation motivates the orientation gate in our pipeline.

B. Normal estimation and surface analysis

A complementary route reasons about local surface geometry rather than global primitives. Huang *et al.* [12] consolidate unorganised clouds and estimate reliable normals for surface reconstruction, whereas Castillo *et al.* [13] recover normals through constrained nonlinear least squares for joint segmentation and denoising. Per-point normals are attractive for stairs because tread and riser differ sharply in orientation. On its own, though, normal estimation produces no grouping into coherent surfaces and remains sensitive to neighbourhood scale under noise. Our PCA-normal baseline builds directly on this principle, classifying points by the tilt of the smallest-eigenvector normal relative to the vertical axis.

C. Clustering and region growing

To recover surfaces rather than isolated points, clustering and region-growing methods aggregate neighbours by geometric similarity. Density-based clustering (DBSCAN) has been used to identify and localise 3D weld seams robustly in the presence of outliers [14]. Euclidean clustering supports object detection from fused sparse clouds and imagery [15], and supervoxel-based region growing yields compact, boundary-respecting segments [16]. Approaches of this kind naturally reject outliers as noise and form one cluster per physical surface, which suits the discrete tread and riser structure of stairs. What they do not encode is the horizontal-versus-vertical distinction we require, so we precede spatial clustering with an orientation split and confirm each cluster by its slope.

D. Ground and LiDAR segmentation

Separating traversable ground from obstacles is a long-studied special case of surface segmentation. The survey by Gomes *et al.* [8] catalogues ground-segmentation methods for automotive LiDAR and highlights a recurring trade-off: geometric methods are fast and transparent, while learned methods are accurate but opaque. Stairs are a structured instance of this ground-versus-non-ground problem, in which each tread is a small, elevated ground patch rather than a single dominant plane.

E. Learning-based segmentation

Deep networks have reshaped point-cloud semantic segmentation since PointNet [6] demonstrated direct learning on raw point sets, and recent surveys document the breadth of this progress [17]. Because dense labels are costly, weakly supervised [18], few-shot [19], and prototype-based few/zero-shot [20] formulations all aim to reduce annotation demand. Accuracy is strong, but these models stay data-hungry, depend on representative training distributions, and offer limited interpretability and reproducibility for a safety-critical sub-problem such as foothold detection. That cost in data and transparency is precisely what motivates revisiting classical geometry for stairs.

F. Robotic stair and terrain perception

In robotics, stair and terrain perception is tightly coupled to control. Matsumura *et al.* [5] detect stairs from 3D point clouds with deep learning to prevent falls, and operator-assisted stair traversal has been demonstrated in field systems [4]. Legged platforms lean on elevation maps and adaptive planning [2], egocentric vision [1], and stereo-based navigation over rough terrain [3], while traversability itself can be learned from simulation [7]. Such systems confirm that reliable steppable-surface detection is a prerequisite for safe motion. They are typically evaluated end-to-end on bespoke platforms, however, which leaves the underlying segmentation step without an isolated, comparable benchmark.

G. Gap addressed by this work

Across these threads, geometric methods are fast and interpretable yet rarely compared head-to-head on stairs, while learned methods require labelled data that is unavailable for this specific task. No prior study offers a controlled, fully labelled staircase benchmark on which classical segmenters can be measured against a common ground truth. We fill this gap with a parametric, fully annotated point-cloud generator and an apples-to-apples comparison of four classical methods for steppable-surface detection.

III. PROPOSED METHOD

This section formalises the steppable-surface segmentation problem. It then describes a parametric generator that produces a fully labelled staircase point cloud with realistic sensor artefacts. Finally, it details four classical geometric methods that we implement and compare under an identical 3-class formulation.

Algorithm 1 Orientation-aware segmentation (generic)

- 1: estimate per-point normals $\hat{\mathbf{n}}_p$ via PCA over k neighbours (4)
- 2: split points into horizontal / vertical orientation groups
- 3: cluster or fit planes within each group
- 4: confirm each region by its plane tilt $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$
- 5: label tread / riser / other

A. Problem Formulation and Class Scheme

We treat staircase perception as a point-wise classification problem. Given an unorganised cloud $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^M$ with $\mathbf{x}_i \in \mathbb{R}^3$, the goal is to assign every point a label $\ell_i \in \{0, 1, 2\}$, where label 0 denotes a *tread* (a steppable, nominally horizontal surface), label 1 denotes a *riser* (the vertical face connecting consecutive treads), and label 2 denotes *other* (structural edges, sensor noise, and outliers). The tread class is the primary class of interest, since a robot or assistive device plans footholds on tread surfaces. We therefore evaluate it in a one-vs-rest setting while still reporting overall 3-class accuracy.

The decisive geometric cue separating the first two classes is surface orientation relative to the vertical axis. Let $\hat{\mathbf{n}}$ be the unit surface normal at a point and $\hat{\mathbf{z}}$ the world vertical. A tread satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \rightarrow 1$ (its normal is nearly parallel to $\hat{\mathbf{z}}$), whereas a riser satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \rightarrow 0$ (its normal is nearly orthogonal to $\hat{\mathbf{z}}$). Every method below exploits this orientation contrast, either explicitly through a normal-based gate or implicitly through the height structure of the cloud. Treating orientation as the organising principle yields the unified pipeline summarised in Algorithm 1. Each of the four methods plugs into this pipeline at the clustering or plane-fitting stage.

B. Parametric Data Generation

To obtain a controlled and fully labelled benchmark, we synthesise the staircase analytically from physical parameters: width $w = 2.5$ m, tread depth $d = 0.30$ m, step height $h = 0.18$ m, and $N = 4$ steps. For step index $i \in \{0, \dots, N-1\}$, tread points are sampled uniformly over the horizontal extent of step i and placed at height ih with additive Gaussian height noise following (1):

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y \sim \mathcal{U}(id, (i+1)d) \\ z = ih + \epsilon_z, \quad \epsilon_z \sim \mathcal{N}(0, \sigma_z^2) \end{cases} \quad (1)$$

$$\begin{cases} x \sim \mathcal{U}(0, w) \\ y = id + \epsilon_y, \quad \epsilon_y \sim \mathcal{N}(0, \sigma_y^2) \\ z \sim \mathcal{U}((i-1)h, ih) \end{cases} \quad (2)$$

Riser points come from (2) for $i \in \{1, \dots, N-1\}$, the vertical face rising to tread i . They share the same lateral spread but lie on a vertical plane at depth id with small along-depth jitter $\sigma_y = 0.02$ m, and their height sweeps uniformly across one step rise. The nominal height noise is $\sigma_z = 0.02$ m,

which we later vary to probe sensitivity. On top of this structured cloud we inject realism in two ways. About 4% of the points are replaced by uniformly distributed outliers spanning the staircase bounding volume, mimicking spurious returns and edge artefacts; these carry the *other* label. Every retained point then receives an isotropic LiDAR-like jitter drawn from $\mathcal{N}(0, 0.005^2)$ on each coordinate, approximating the few-millimetre range noise typical of indoor depth sensors. We treat this value as a nominal modelling choice, not a calibrated sensor model, and probe its principal axis (σ_z) directly in Section IV. The resulting cloud contains 18,333 points, partitioned as 12,800 tread, 4,800 riser, and 733 other. The generator emits per-point labels by construction, so it provides exact ground truth for both training-free evaluation and the noise sweep. This removes the manual-annotation bottleneck that hampers learned segmenters.

C. RANSAC Plane Fitting

The first method fits planes by Random Sample Consensus. A candidate plane with unit normal $\hat{\mathbf{n}}$ and offset b scores points by their orthogonal distance. The inlier set $\mathcal{I}$ then collects every point closer than a tolerance τ :

$$d(\mathbf{x}) = |\hat{\mathbf{n}}^\top \mathbf{x} + b|, \quad \mathcal{I} = \{\mathbf{x} : d(\mathbf{x}) < \tau\} \quad (3)$$

A naive single fit fails on stairs. The dominant plane that maximises inliers is the oblique surface spanning the entire staircase, whose normal has $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| = d/\sqrt{d^2 + h^2} \approx 0.86$. RANSAC therefore returns a spurious “ramp” that absorbs both treads and risers into one steppable region. We avoid this with a two-phase, orientation-gated scheme. The first phase extracts horizontal planes (treads) by accepting only fits whose normal satisfies $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \geq 0.93$. The second reruns RANSAC on the residual points to extract vertical planes (risers) with $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}| \leq 0.25$. We set $\tau = 0.035$ m and impose minimum inlier counts of 2,000 points for tread planes and 300 for riser planes. This suppresses thin riser slices from contaminating tread fits, at the cost of occasionally rejecting sparse vertical faces. The orientation-aware variant follows the spirit of structure-guided RANSAC refinements [9], [10].

D. PCA Normal Analysis

The second method classifies each point purely from its local surface orientation. For point p we gather its k nearest neighbours $\mathcal{N}_k(p)$ and form the local covariance matrix $\mathbf{C}_p$. The eigenvector tied to its smallest eigenvalue serves as the surface normal:

$$\mathbf{C}_p = \frac{1}{k} \sum_{j \in \mathcal{N}_k(p)} (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top, \quad \hat{\mathbf{n}}_p = \mathbf{v}_{\min}(\mathbf{C}_p) \quad (4)$$

The smallest-eigenvalue direction is the axis of least point spread, which for a locally planar patch coincides with its normal [12], [13]. With $k = 50$ the estimate stays stable against the injected jitter while remaining local enough to respect tread and riser boundaries. We classify by the verticality of the estimated normal: a point is labelled tread when $|\hat{\mathbf{n}}_p^\top \hat{\mathbf{z}}| \geq 0.80$

and riser when $|\hat{\mathbf{n}}_p^\top \hat{\mathbf{z}}| \leq 0.50$. Intermediate values, which arise mostly at edges and from outliers, fall to the *other* class. This decision is per-point and parameter-light, with no plane-fitting iterations. The PCA thresholds (0.80/0.50) are deliberately looser than the RANSAC orientation gate (0.93/0.25). RANSAC tests a global plane whose aggregate normal is precise, whereas per-point PCA normals are noisier and need a wider acceptance band.

E. DBSCAN with Slope Confirmation

The third method couples orientation reasoning with spatial clustering. We first pre-split the cloud into a horizontal group and a vertical group using the same normal-verticality criterion as above. Within each group we run density-based spatial clustering (DBSCAN) with neighbourhood radius $\varepsilon = 0.08$ m, so that each physically separate surface forms its own connected cluster, while sparse outliers, lacking dense support, fall to the noise label. Pre-splitting by orientation matters here. Without it, a tread and the riser directly beneath it, which are spatially adjacent, would merge into a single density-connected cluster. Each resulting cluster is confirmed by its aggregate slope, that is the verticality $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$ of its fitted plane, and labelled tread, riser, or other accordingly. This pairing of clustering with region-level geometric confirmation mirrors established point-cloud clustering and region-growing practice [14], [16].

F. Height-Histogram Analysis

The fourth method is the most specialised to the staircase structure and ignores normals entirely. Every tread of an ideal staircase lies at a discrete height ih , so the marginal distribution of the z coordinate concentrates into sharp peaks at those levels, separated by low-density bands that correspond to the riser sweeps. We build a one-dimensional histogram of z over the whole cloud and detect its dominant peaks. Points falling within a window of half-width 0.05 m around each peak are labelled tread, points in the intervening height bands are labelled riser, and the remainder become other. On the benchmark cloud this recovers four tread levels at $z \approx \{0.00, 0.18, 0.36, 0.54\}$ m, matching the generative parameters. The method is extremely fast and needs no neighbourhood computation, but it presumes a regular, axis-aligned staircase. As the evaluation shows, its height bands tend to over-include riser points near each tread level.

G. Implementation and Parameters

All methods were implemented in Python with NumPy and scikit-learn. RANSAC and the normal-based gates are custom, whereas DBSCAN and the nearest-neighbour search reuse standard library routines. Table I consolidates every parameter for reproducibility. The four methods share an asymptotic cost dominated by the k -nearest-neighbour normal estimation ($O(M \log M)$ with a k -d tree for M points). On top of this, RANSAC adds a term linear in the iteration budget, DBSCAN a near-linear neighbourhood query, and the height-histogram stays at $O(M)$, which makes it by far the cheapest. All values were held fixed across the experiments reported in Section IV.

TABLE I
HYPER-PARAMETERS HELD FIXED ACROSS ALL EXPERIMENTS.

Symbol	Value	Description
w, d, h	2.5, 0.30, 0.18 m	width, tread depth, step height
N	4	number of steps
σ_z, σ_y	0.02, 0.02 m	tread / riser Gaussian noise
jitter	0.005 m	isotropic LiDAR-like jitter
outliers	4%	uniform outlier fraction
τ	0.035 m	RANSAC inlier threshold
min. inliers	2,000 / 300	RANSAC tread / riser plane
k	50	PCA normal neighbours
gate (RANSAC)	0.93 / 0.25	horiz. / vert. $ \hat{\mathbf{n}}^\top \hat{\mathbf{z}} $
gate (PCA)	0.80 / 0.50	tread / riser $ \hat{\mathbf{n}}^\top \hat{\mathbf{z}} $
ε	0.08 m	DBSCAN radius
window	0.05 m	histogram peak half-width

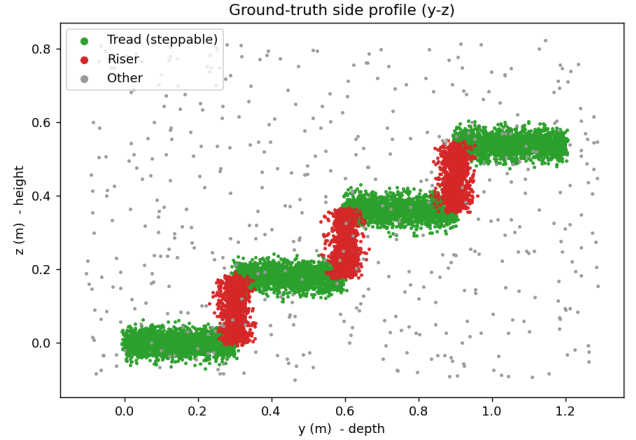


Fig. 1. Side profile (y - z) of the ground-truth cloud: green = tread, red = riser, grey = other.

IV. EVALUATION

This section compares the four classical segmentation methods on the parametric staircase point cloud. The setup and metrics come first (A), followed by the quantitative results and their interpretation (B). We then examine robustness as height noise grows (C) and close with the safety implications for stair-following robots (D).

A. Experimental Setup and Metrics

All experiments ran on the synthetic staircase from Section III, with $N = 4$ steps, width $w = 2.5$ m, tread depth $d = 0.30$ m, and step height $h = 0.18$ m. The generated cloud holds 18,333 points: 12,800 tread points (label 0), 4,800 riser points (label 1), and 733 “other” points (label 2) that come from edges, LiDAR jitter, and the injected outliers. Figure 1 shows the y - z side profile of the ground-truth cloud. The alternating horizontal treads and vertical risers stand out clearly, along with the sparse off-surface clutter.

The steppable surface is the safety-critical class, so we score every method on the tread label in a one-vs-rest setting: tread points are positives, and all riser and “other” points are negatives. Let TP , FP , FN , and TN denote the per-point true positives, false positives, false negatives, and true negatives.

TABLE II

TREAD-CLASS SEGMENTATION METRICS (ONE-VS-REST). ACC IS ONE-VS-REST TREAD ACCURACY, DISTINCT FROM THE THREE-CLASS ACCURACY QUOTED IN THE TEXT. BEST PER COLUMN IN BOLD.

Method	Acc	Prec	Rec	F1	IoU
RANSAC	0.681	0.714	0.904	0.798	0.664
PCA-Normals	0.901	0.954	0.901	0.927	0.864
DBSCAN-Slope	0.920	0.936	0.949	0.943	0.892
Height-Histogram	0.827	0.809	0.984	0.888	0.799

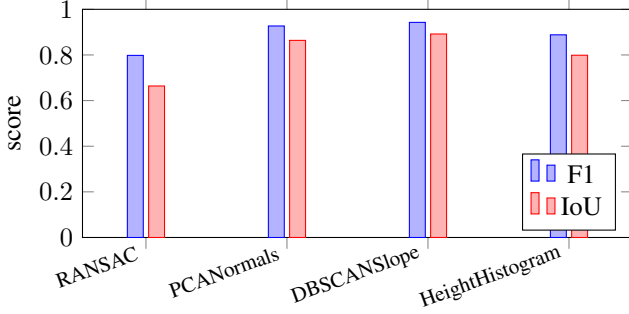


Fig. 2. Per-method tread-class F1 and IoU.

We report five standard quantities, each computed for the tread class under this one-vs-rest scheme. The tabulated accuracy $\text{Acc} = (TP+TN)/(TP+TN+FP+FN)$ is the fraction of points placed correctly inside or outside the tread class. In the text we also quote the overall *three-class* accuracy, the fraction of all points whose full tread/riser/other label is correct; this differs from the one-vs-rest Acc of Table II. Precision $P = TP/(TP+FP)$ is the fraction of predicted tread points that are genuinely treads, while recall $R = TP/(TP+FN)$ is the fraction of true tread points recovered. The F1 score is their harmonic mean, $F1 = 2PR/(P+R)$, and the intersection-over-union, or Jaccard index, $\text{IoU} = TP/(TP+FP+FN)$, expresses the spatial overlap between the predicted and ground-truth tread regions. Precision and IoU matter most for robotic foothold reasoning, because they directly measure how often a non-steppable surface is taken for a steppable one.

Reproducibility was a design goal. The data generator and every stochastic component of the pipeline (RANSAC sampling, neighbour selection, and the noise draws) ran from a fixed pseudo-random seed, so the cloud and all reported numbers regenerate deterministically. Method hyper-parameters stayed constant across runs: RANSAC used an inlier threshold $\tau = 0.035\text{m}$, PCA normals were estimated from $k = 50$ neighbours, and DBSCAN used $\varepsilon = 0.08\text{m}$ within each orientation group.

B. Results

Table II summarises the tread-class performance of the four methods, and Fig. 2 places their F1 and IoU scores side by side.

The orientation-aware **DBSCAN-Slope** method leads overall. It attains the best F1 (0.943) and IoU (0.892) and the

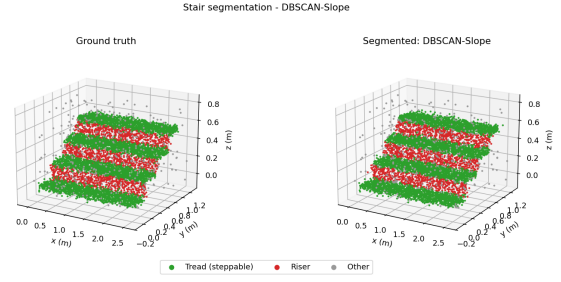


Fig. 3. 3D segmentation by DBSCAN with slope analysis (right) vs. ground truth (left).

highest three-class accuracy (0.909). The reason is its order of operations: by first splitting the cloud into horizontal and vertical orientation groups and clustering spatially only afterwards, it recovers each physical surface as a single compact cluster and absorbs outliers into the noise label. The result is a balanced precision (0.936) and recall (0.949). Figure 3 sets the DBSCAN-Slope segmentation against the ground truth; the individual treads separate cleanly, and riser bleed-through is minimal.

The **PCA-Normals** method follows close behind, with F1 0.927 and IoU 0.864. It classifies points directly from the local surface orientation $|\hat{\mathbf{n}}^\top \hat{\mathbf{z}}|$, and this earns it the highest precision of any method (0.954): a point is labelled tread only when its estimated normal is nearly vertical, so very few risers slip through. Recall (0.901) sits marginally below DBSCAN's, because normals estimated near tread-riser edges grow unreliable and a fraction of genuine tread points fall outside the horizontal gate. The narrow gap between PCA-Normals and DBSCAN-Slope confirms that local geometric orientation is an effective, robust cue for separating the two principal surface types.

The **Height-Histogram** method acts as a high-recall, lower-precision detector. It reaches the best recall of any method (0.984) but a markedly lower precision (0.809), giving F1 0.888 and IoU 0.799. Figure 4 shows the z -histogram, whose four pronounced peaks correspond to the detected tread levels at $z \approx \{0.00, 0.18, 0.36, 0.54\}\text{m}$, close to the nominal step heights. The method captures almost every tread point because treads concentrate sharply at these levels. The trouble is that the horizontal bands it assigns around each peak also enclose riser points whose vertical extent happens to pass through a tread height, and that drags precision down.

Plain **RANSAC** plane fitting is the weakest performer. It posts the lowest precision (0.714), F1 (0.798), IoU (0.664), and three-class accuracy (0.667), even with a high recall (0.904). One failure mode dominates: the large horizontal planes RANSAC fits at each tread height also swallow riser points that lie within the inlier band τ at those heights. These mislabelled risers inflate the false-positive count and collapse precision. The contrast with PCA-Normals is telling. A global planar fit cannot tell a riser point that merely sits near a tread plane from a true tread point, whereas a local normal can. This is why the per-point orientation cues in PCA-Normals

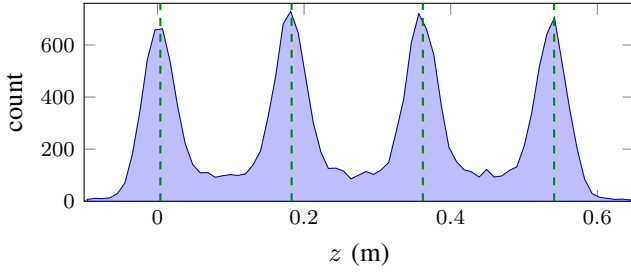


Fig. 4. Height histogram of z ; dashed lines mark the four detected tread levels.

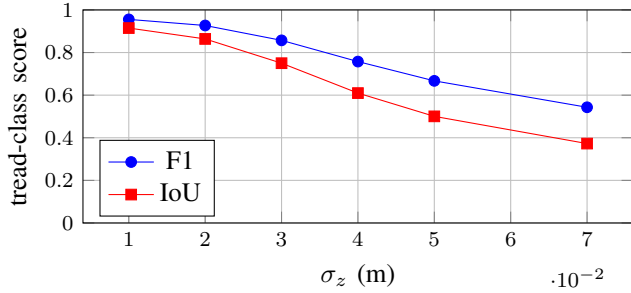


Fig. 5. Tread-class F1 and IoU versus height noise σ_z (PCA-Normals).

and DBSCAN-Slope beat the purely planar formulation.

C. Noise Sensitivity

To probe robustness, we swept the height-noise standard deviation σ_z from 0.01 m to 0.07 m and re-evaluated PCA-Normals, whose accuracy hinges most directly on the fidelity of local geometry. We vary σ_z rather than the riser depth jitter σ_y because vertical jitter is the axis that erodes the tread/riser orientation contrast, whereas σ_y mostly broadens riser planes without moving tread points across step levels. Figure 5 plots the resulting tread-class F1 and IoU.

Performance degrades monotonically, and the slope steepens as the noise grows. At $\sigma_z = 0.01$ m the method is near-ideal (F1 0.956, IoU 0.915), and it holds up well through $\sigma_z = 0.02$ – 0.03 m. Past that point the decline accelerates, reaching F1 0.543 and IoU 0.373 at $\sigma_z = 0.07$ m. The mechanism is geometric. The discriminative cue is the near-vertical tread normal, but once σ_z approaches half the step height ($h/2 = 0.09$ m), vertical jitter on tread points becomes comparable to the vertical gap between adjacent steps. The local neighbourhoods used for normal estimation then mix points from neighbouring treads and the intervening riser, the estimated normals tilt away from vertical, the horizontal orientation gate misfires, and tread and riser turn geometrically indistinguishable. This marks a practical operating limit: reliable tread segmentation needs the sensor’s per-point height noise to stay well below the step height it must resolve.

D. Limitations and Threats to Validity

Several limitations bound these conclusions. The benchmark is synthetic, a single straight-flight, rectangular staircase with

one fixed geometry (w, d, h, N) and an idealised, axis- and gravity-aligned layout. The absolute scores should therefore be read as *relative* method rankings, not field-deployable accuracies. The sensor model is also simplified: nominal Gaussian jitter plus uniform outliers, with no occlusion, missing returns, incidence-angle effects, multi-path, or the structured scan patterns of a real LiDAR. The height-histogram method, for its part, assumes a regular, gravity-aligned staircase and would not transfer to spiral, open-riser, or unaligned scans without modification. Finally, every hyper-parameter (τ, k, ε , the orientation gates, and the histogram window) was tuned once on this cloud, and the sensitivity study varies only σ_z for a single method, so we make no claim of cross-configuration parameter robustness. These constraints motivate the real-sensor validation identified in Section V.

E. Safety Discussion for Stair-Following Robots

For a stair-climbing or stair-following robot, the worth of these results lies less in the average score than in the failure modes they expose. The best method reaches IoU ≈ 0.89 , enough to support *coarse* foothold planning: the extracted tread regions overlap the true steppable surfaces strongly enough for an elevation-map-based planner to place candidate footholds on the correct steps, in line with practice in legged and wheeled terrain perception [7], [8]. Yet no method reaches unit precision, and the dominant errors are exactly the dangerous ones: riser points occasionally read as steppable. Inside a closed control loop such false positives can lead a controller to commit a foothold to a vertical face, the very failure mode that motivates dedicated stair perception [5].

We therefore recommend that a deployed system treat the raw per-point labels as a hypothesis rather than a command. Three mitigations follow directly from our findings. The first is *ensemble voting* across methods. Because the methods fail in complementary ways (RANSAC and the histogram over-include risers, while PCA-Normals stays conservative), voting can suppress method-specific false positives and combine the histogram’s high recall with the precision of the normal-based gate. The second is a *tread-edge erosion margin*. Mislabelling concentrates at tread–riser boundaries, so shrinking each accepted tread region inward before foothold selection trades a small amount of usable surface for a sharp drop in the chance of stepping onto an edge or riser. The third is *temporal verification*. Given the strong noise sensitivity established above, accumulating evidence across consecutive scans before committing a foothold guards against transient mislabels caused by momentary high-noise frames. Together these measures convert a segmentation that is accurate on average into one whose residual errors lean toward the safe side, which is the right design objective for robots that physically traverse stairs.

V. CONCLUSION

We presented a parametric, fully labeled staircase point-cloud benchmark together with an apples-to-apples comparison of four classical geometric methods: RANSAC plane

fitting, PCA normal analysis, DBSCAN with slope confirmation, and height-histogram banding. All four operate under a unified three-class formulation of treads, risers, and other points. On the steppable tread class, orientation-aware DBSCAN with per-cluster slope confirmation was the strongest method, reaching an F1 of 0.943 and an IoU of 0.892, with PCA-normals a close second. These findings rest on a single synthetic staircase under a controlled noise model, so they should be read as relative method rankings pending real-sensor validation. RANSAC was the weakest, primarily because horizontal planes fitted at tread heights absorb riser points and depress precision to 0.714; the height-histogram, by contrast, attained the highest recall but conflated riser bands with treads. A noise-sensitivity sweep on the PCA-Normals method showed that its performance holds up at realistic jitter but degrades sharply once the height noise approaches half the step height, with tread F1 falling from 0.956 at $\sigma_z = 0.01$ m to 0.543 at $\sigma_z = 0.07$ m.

From a safety standpoint, an IoU near 0.89 is adequate for coarse foothold planning. Yet no method reached unit precision, so residual mislabeling of risers as steppable remains a hazard for stair-following robots and assistive devices, which argues for conservative tread-edge margins and downstream verification. Future work will pursue ensemble fusion of the complementary geometric cues to recover precision, validation on real LiDAR staircases beyond the synthetic generator, and a controlled comparison against learned point-cloud segmenters [6], [17] under low-label regimes, where the reproducibility and interpretability of classical methods matter most.

REFERENCES

- [1] A. Agarwal, A. Kumar, J. Malik, and D. Pathak, "Legged locomotion in challenging terrains using egocentric vision," *Conference on Robot Learning*, 2022. [Online]. Available: <https://doi.org/10.48550/arXiv.2211.07638>
- [2] D. Belter, P. Labecki, and P. Skrzypczyński, "Adaptive motion planning for autonomous rough terrain traversal with a walking robot," *J. Field Robotics*, 2016. [Online]. Available: <https://doi.org/10.1002/rob.21610>
- [3] A. Stelzer, H. Hirschmüller, and M. Görner, "Stereo-vision-based navigation of a six-legged walking robot in unknown rough terrain," *Int. J. Robotics Res.*, 2012. [Online]. Available: <https://doi.org/10.1177/0278364911435161>
- [4] P. Marion, M. Fallon, R. Deits, A. K. Valenzuela, C. Pérez-D'Arpino, G. Izatt, L. Manuelli, M. E. Antone, H. Dai, T. Koolen, J. Carter, S. Kuindersma, and R. Tedrake, "Director: A user interface designed for robot operation with shared autonomy," *J. Field Robotics*, 2017. [Online]. Available: <https://doi.org/10.1002/rob.21681>
- [5] H. Matsumura and C. Premachandra, "Deep-learning-based stair detection using 3d point cloud data for preventing walking accidents of the visually impaired," *IEEE Access*, 2022. [Online]. Available: <https://doi.org/10.1109/ACCESS.2022.3178154>
- [6] C. Qi, H. Su, K. Mo, and L. Guibas, "Pointnet: Deep learning on point sets for 3d classification and segmentation," *Computer Vision and Pattern Recognition*, 2016. [Online]. Available: <https://doi.org/10.1109/CVPR.2017.16>
- [7] R. O. Chavez-Garcia, J. Guzzi, L. Gambardella, and A. Giusti, "Learning ground traversability from simulations," *IEEE Robotics and Automation Letters*, 2017. [Online]. Available: <https://doi.org/10.1109/LRA.2018.2801794>
- [8] T. Gomes, D. Matias, A. N. Campos, L. Cunha, and R. Roriz, "A survey on ground segmentation methods for automotive lidar sensors," *Italian National Conference on Sensors*, 2023. [Online]. Available: <https://doi.org/10.3390/s23020601>
- [9] L. Li, F. Yang, H. Zhu, D. Li, Y. Li, and L. Tang, "An improved ransac for 3d point cloud plane segmentation based on normal distribution transformation cells," *Remote Sensing*, 2017. [Online]. Available: <https://doi.org/10.3390/rs9050433>
- [10] A. Jalal, M. Sarwar, and K. Kim, "Rgb-d images for objects recognition using 3d point clouds and ransac plane fitting," in *International Bhurban Conference on Applied Sciences and Technology*, 2021. [Online]. Available: <https://doi.org/10.1109/IBCAST51254.2021.9393166>
- [11] L. Zhou, S. Wang, and M. Kaess, " π -lsam: Lidar smoothing and mapping with planes," *IEEE International Conference on Robotics and Automation*, 2021. [Online]. Available: <https://doi.org/10.1109/ICRA48506.2021.9561933>
- [12] H. Huang, D. Li, H. Zhang, U. Ascher, and D. Cohen-Or, "Consolidation of unorganized point clouds for surface reconstruction," *ACM Transactions on Graphics*, 2009. [Online]. Available: <https://doi.org/10.1145/1661412.1618522>
- [13] E. Castillo, J. Liang, and H. Zhao, "Point cloud segmentation and denoising via constrained nonlinear least squares normal estimates," *Innovations for Shape Analysis, Models and Algorithms*, 2013. [Online]. Available: https://doi.org/10.1007/978-3-642-34141-0_13
- [14] N. Zhou, P. Jiang, S. Jiang, L. Shu, X. Ni, and L. Zhong, "An identification and localization method for 3d workpiece welds based on the dbscan point cloud clustering algorithm," *Journal of Manufacturing and Materials Processing*, 2024. [Online]. Available: <https://doi.org/10.3390/jmmp8060287>
- [15] X. Xu, L. Zhang, J. Yang, C. Cao, Z. Tan, and M. Luo, "Object detection based on fusion of sparse point cloud and image information," *IEEE Transactions on Instrumentation and Measurement*, 2021. [Online]. Available: <https://doi.org/10.1109/TIM.2021.3102739>
- [16] N. Luo, Y. Jiang, and Q. Wang, "Supervoxel-based region growing segmentation for point cloud data," *International journal of pattern recognition and artificial intelligence*, 2021. [Online]. Available: <https://doi.org/10.1142/s0218001421540070>
- [17] S. Sarker, P. Sarker, G. Stone, R. Gorman, A. Tavakkoli, G. Bebis, and J. Sattarvand, "A comprehensive overview of deep learning techniques for 3d point cloud classification and semantic segmentation," *Machine Vision and Applications*, 2024. [Online]. Available: <https://doi.org/10.1007/s00138-024-01543-1>
- [18] M. Li, Y. Xie, Y. Shen, B. Ke, R. Qiao, B. Ren, S. Lin, and L. Ma, "Hybridcr: Weakly-supervised 3d point cloud semantic segmentation via hybrid contrastive regularization," *Computer Vision and Pattern Recognition*, 2022. [Online]. Available: <https://doi.org/10.1109/CVPR52688.2022.01451>
- [19] N. Zhao, T.-S. Chua, and G. H. Lee, "Few-shot 3d point cloud semantic segmentation," *Computer Vision and Pattern Recognition*, 2020. [Online]. Available: <https://doi.org/10.1109/CVPR46437.2021.00876>
- [20] S. He, X. Jiang, W. Jiang, and H. Ding, "Prototype adaption and projection for few- and zero-shot 3d point cloud semantic segmentation," *IEEE Transactions on Image Processing*, 2023. [Online]. Available: <https://doi.org/10.1109/TIP.2023.3279660>